

# Enhancing Domain-Adaptive Hashing via Evidential Learning and Progressive Alignment

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## Abstract

Domain-adaptive hashing enhances discriminative hash representations by transferring knowledge from a label-rich source domain to a label-scarce target domain. It has attracted significant attention due to its ability to enable efficient cross-domain retrieval without requiring target domain labels. However, existing methods generally assume that source domain labels are completely accurate. In practice, labels obtained via web crawling or crowdsourcing often contain varying degrees of noise, which hampers semantic alignment and aggravates domain shift. To tackle these issues, we propose a novel method termed **Evidential Learning and Progressive Alignment (ELPA)** for domain-adaptive hashing. This method comprises two key modules: the Uncertainty-aware Noise Separation (UNS) and the Progressive Cross-domain Alignment (PCA). In the UNS, we exploit the belief and uncertainty masses obtained from the evidential learning model and utilize the posterior probabilities of a Gaussian Mixture Model to effectively distinguish clean samples from noisy ones. In PCA, we introduce a progressive partial optimal transport mechanism that prioritizes pseudo-label generation for well-aligned target samples, thereby gradually achieving class-level and global-level cross-domain alignment. Extensive experiments across multiple benchmark datasets with various noise ratios demonstrate that ELPA consistently surpasses existing state-of-the-art methods, exhibiting superior robustness and generalization capability.

## CCS Concepts

• Information Systems → Information Search.

## Keywords

Domain-adaptive hashing, Noisy labels, Evidential learning, Progressive alignment

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## 1 Introduction

With the explosive growth of large-scale multimodal data, achieving efficient and accurate similarity retrieval has become a key research challenge. Hash-based approximate nearest neighbor (ANN) search methods aim to significantly improve the efficiency of similarity retrieval while maintaining retrieval accuracy by mapping high-dimensional features into compact binary hash spaces. Owing to their high efficiency and low storage cost, these methods have been widely applied across various fields, such as cross-modal retrieval [31, 46, 47], retrieval-augmented generation (RAG) [10, 44], e-commerce recommendation [3, 15], and face recognition [32, 33].

Domain-adaptive (DA) hash retrieval aims to use image data from one domain as queries to retrieve semantically related samples from another domain. Early DA hashing methods combined cross-domain similarity preservation techniques [24, 35] with domain adaptation techniques [11, 12] to generate domain-invariant hash codes. In recent years, with the rapid development of deep learning, a series of deep DA hashing approaches have emerged. Most of these approaches [25, 37] follow general domain adaptation paradigms, yet struggle to effectively integrate domain adaptation with hash learning. Although they have achieved remarkable progress in domain-adaptive hash retrieval, their performance remains constrained by the purity of the labeled source data. For instance, when using photos captured by mobile devices to retrieve corresponding product images from an e-commerce platform, the latter often suffers from noisy labels due to inaccurate user annotations or automated web crawling, which undermines the reliability of cross-domain retrieval. Therefore, this paper explores the more challenging and practically significant problem of domain-adaptive hashing retrieval under noisy conditions.

Currently, noise-robust domain-adaptive hashing faces two major unresolved challenges that hinder further performance improvements. 1) No effective noise modeling mechanism. Noisy labels in the source domain propagate incorrect semantic information, which not only amplifies domain shift but also causes misalignment

of hash codes in the Hamming space. Therefore, how to effectively model and correct source domain noise in domain-adaptive hashing remains an open and underexplored problem. 2) Insufficient domain alignment under noisy conditions. Most existing approaches perform cross-domain alignment through adversarial learning [24] or instance/prototype-level contrastive learning [4, 38]. However, their effectiveness largely depends on the quality of pseudo-labels generated in the target domain. Bias and overconfidence in these pseudo-labels can lead to the accumulation of erroneous semantics during training, thereby weakening the discriminability of the learned hash codes. Thus, how to avoid pseudo-label bias and achieve robust cross-domain alignment under noisy conditions remains a key challenge to be addressed in domain-adaptive hashing.

To address the aforementioned challenges, we propose a simple yet effective method, termed Evidential Learning and Progressive Alignment (ELPA) for domain-adaptive hashing, as illustrated in Figure 1. This method leverages the model’s inherent ability to quantify sample prediction uncertainty under the evidence-based learning paradigm to distinguish clean samples from noisy ones, thereby facilitating targeted correction or suppression for noisy samples. More specifically, we first calculate the belief mass and uncertainty mass of each sample and simultaneously fit the evidential loss of each sample using a Gaussian mixture model. Then, based on the joint constraints of posterior probability and uncertainty mass, we perform collaborative partitioning to divide source data into clean and noisy subsets. For noisy samples, we further design a noise-resilient label correction strategy that leverages the semantic information of their corresponding high-confidence clean neighbors to correct them in the feature space, thereby enhancing both distinguishability and semantic consistency. Finally, to address the pseudo-label noise issue in cross-domain alignment, we propose a progressive partial optimal transport module. This module prioritizes generating pseudo-labels for well-aligned target samples, thereby enhancing both class-level and global-level alignment, and ultimately improving the robustness and transferability of the learned hash representations.

The key contributions of our work can be summarized as follows:

- We investigate the noise issue in domain-adaptive hash retrieval and propose a novel Evidential Learning and Progressive Alignment (ELPA) method.
- To address the issue of insufficient source domain noise modeling, we propose an uncertainty-aware noise separation strategy. Combined with uncertainty-guided collaborative partitioning, this approach can more effectively separate noisy and clean samples.
- To alleviate the problem of target domain pseudo-label noise in cross-domain alignment, we propose a progressive partial optimal transport method. By selecting reliable target samples to generate pseudo-labels, we achieve more robust class-level and global-level feature alignment.
- Extensive experimental results demonstrate that our ELPA consistently outperforms state-of-the-art domain-adaptive hashing methods across various levels of label noise, achieving significant improvements on key metrics.

## 2 Related Work

### 2.1 Domain-Adaptive Hashing

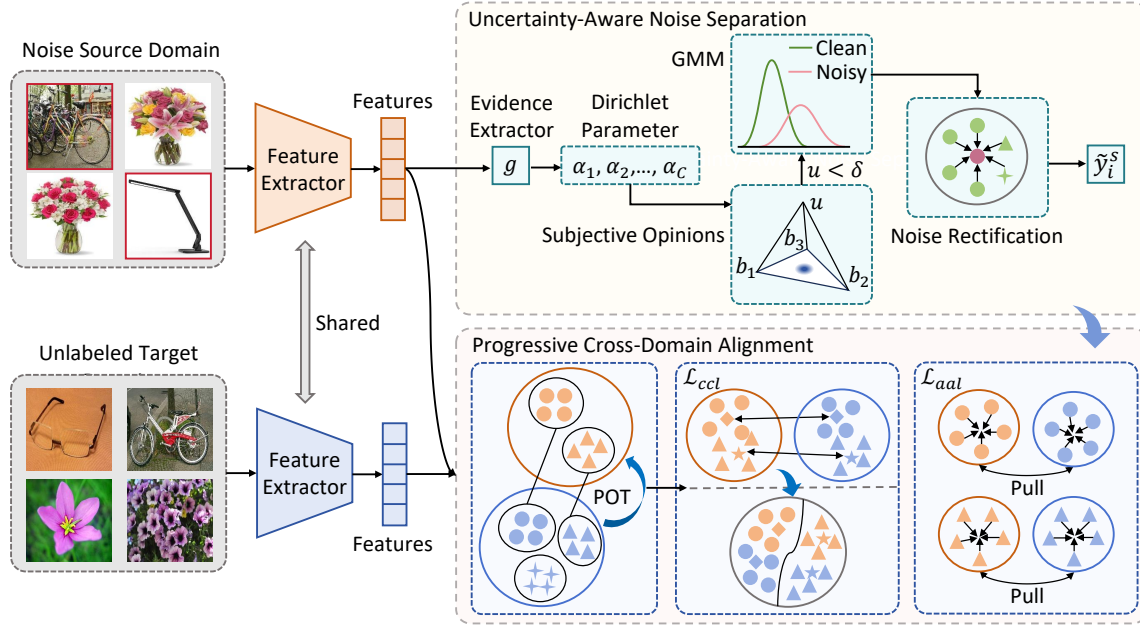
Domain-Adaptive Hashing (DAH) was developed to retrieve relevant samples from different domains using image data from one domain as a query. DAH [35] is the first work in this field. It jointly optimizes supervised hash loss, unsupervised entropy loss for the target domain, and multi-kernel Maximum Mean Discrepancy (MK-MMD) loss to align features between the source and target domains. PEACE [37] leverages source labeled data and target unlabeled data, measuring and progressively optimizing the uncertainty of target pseudo-labels to achieve reliable semantic alignment for cross-domain image retrieval. CPH [4] proposes a domain-shared unit hypersphere space through prototype contrastive learning and maps it to the Hamming hypersphere space for cross-domain hash learning. ROSE [42] proposes relational and prototypical structure learning to handle the universal domain adaptation problem. COUPLE [25] uses neighborhood graphs and graph flow diffusion to achieve noise-robust target domain learning and progressive cross-domain alignment.

### 2.2 Uncertainty-based Learning

Uncertainty learning aims to assess the confidence and reliability of deep neural network (DNN) predictions. Based on the sources of uncertainty, uncertainty learning is generally categorized into data uncertainty and model uncertainty. Early research on model uncertainty primarily relied on Bayesian neural networks (BNNs) [16], characterizing parameter-level uncertainty by inferring the posterior distribution of model parameters. To reduce computational and memory overhead, the Monte Carlo (MC) Dropout method [6] estimates the prediction variance by using different dropout masks (i.e., randomly turning off some neurons) in multiple random forward propagations, thereby approximating model uncertainty. In addition, the Ensemble method [18] forms an output distribution by combining multiple independently trained neural networks and uses the variance of the distribution to quantify the uncertainty of the model. To tackle the over-confident false prediction [9], Evidential Deep Learning (EDL) models predict uncertainty by applying Subjective Logic (SL) [17] to neural networks.

### 2.3 Learning with Noisy Labels

Mislabeled data can lead to deep neural networks overfitting to noisy labels. To address the issue of learning from noisy labels, previous researchers have proposed various methods, including noise robust neural network architectures [2], sample selection methods [21, 40, 41], robust regularization techniques [23, 45], meta-learning approaches [27, 28], and robust loss methods [7, 39, 43]. Noise-robust neural network architectures introduce noise adaptation layers or specialized modules in the network to model label transitions. Sample selection methods select clean samples based on loss, confidence, or uncertainty, and train the network only on these samples while excluding noisy ones. Robust regularization techniques use regularization strategies like dropout, weight decay, or data augmentation to reduce overfitting to noisy labels. Meta-learning approaches automatically adjust the training loss or sample weights via meta-learning. Robust loss methods design



**Figure 1: The overview of ELPA.** ELPA comprises two modules: **Uncertainty-Aware Noise Separation** and **Progressive Cross-Domain Alignment**. The first module estimates belief and uncertainty masses to collaboratively separate clean and noisy samples through a GMM guided by posterior probabilities. The second module employs progressive partial optimal transport to generate reliable pseudo-labels for well-aligned target samples, facilitating class-level contrastive and adversarial alignment.

noise-tolerant or reweighted loss functions (e.g., MAE, GCE, symmetric cross-entropy) to mitigate the negative impact of incorrect labels.

### 3 Methodology

#### 3.1 Problem Definition

For clarity, we define the source domain as  $\mathcal{D}_s = \{x_i^s, y_i^s\}_{i=1}^{N_s}$  with noisy labels  $y_i^s$ , and  $\mathcal{D}_t = \{x_i^t\}_{i=1}^{N_t}$  as the unlabeled target domain.  $N_s$  and  $N_t$  denote the total numbers of training samples in the source and target domains, respectively. The source domain  $\mathcal{D}_s$  and the target domain  $\mathcal{D}_t$  originate from the same label space  $Y = \{1, 2, \dots, C\}$  yet have different data distributions, thereby resulting in potential domain shift. Our objective is to learn compact  $L$ -bit hash codes  $B \in \{-1, 1\}^L$  that effectively preserve semantic information. For single-domain retrieval, both the queries and the database are drawn from the target domain  $\mathcal{D}_t$ . In contrast, cross-domain retrieval aims to search for semantically similar samples in the source domain  $\mathcal{D}_s$  given queries from the target domain  $\mathcal{D}_t$ . This setting is more challenging since noisy labels in the source domain not only distort semantic consistency but also exacerbate overfitting, making effective cross-domain alignment more difficult.

#### 3.2 Uncertainty-Aware Noise Separation

**Uncertainty Modeling.** The primary obstacle in cross-domain hashing retrieval with noisy labels is the effective identification of clean versus noisy samples, as they require different processing strategies. To achieve this, we first employ the Dempster-Shafer Theory [34] of evidential learning to characterize the uncertainty

of the source-domain model, which is subsequently utilized as the criterion for partitioning clean data from noisy data. Specifically, for a given source-domain sample  $x_i^s \in \mathcal{D}_s$ , the evidence  $e_i$  is extracted through the evidence extractor, formulated as:

$$e_i = g(f(c|x_i^s)), \quad (1)$$

where  $c \in \{1, 2, \dots, C\}$  indicates the category of the sample,  $f(\cdot)$  is the source domain feature extractor, and  $g(\cdot)$  represents the ReLU activation function.

We then employ Subjective Logic [17] to model the uncertainty in the source domain. Subjective Logic attempts to assign a belief mass to each query and to quantify the uncertainty mass derived from the collected evidence. The belief mass and the uncertainty are defined as follows:

$$b_{ij} = \frac{e_{ij}}{L_i} = \frac{\alpha_{ij} - 1}{L_i} \quad \text{and} \quad u_i = \frac{C}{L_i}, \quad (2)$$

where  $L_i = \sum_{j=1}^C (e_{ij} + 1)$  and  $u + \sum_{j=1}^C b_{ij} = 1$ .  $L_i$  denotes the Dirichlet distribution strength, and belief mass  $b_i = [b_{i1}, b_{i2}, \dots, b_{iC}]$  represents subjective opinions corresponding to the Dirichlet distribution. The parameter  $\alpha_i = [\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iC}]$  of the Dirichlet distribution is calculated by  $\alpha_i = e_i + 1$ , and  $u$  indicates the uncertainty.

The Dirichlet distribution, parameterized by evidence  $e_i$ , describes the density of probability assignments, with  $\alpha_i$  modeling second-order probabilities and uncertainty. The Dirichlet distribution has the following probability density function:

$$D(p_i|\alpha_i) = \begin{cases} \frac{1}{B(\alpha_i)} \prod_{c=1}^C p_{ic}^{\alpha_{ic}-1} & \text{for } p_i \in S_C, \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

where  $p_i$  represents the query probabilities,  $B(\alpha_i)$  denotes the  $C$ -dimensional multinomial beta function, and  $S_C$  denotes the  $C$ -dimensional unit simplex.

**Evidential Learning for Noise Separation (ENS).** To ensure that the query probability  $p_i$  approximates the ground truth  $y_i^s$ , we adopt the following least-squares (LS) loss function [30]:

$$\begin{aligned}\mathcal{L}_{ls}(\alpha_i, y_i^s) &= \int \|y_i^s - p_i\|_2^2 \frac{1}{B(\alpha_i)} \prod_{c=1}^C p_{ic}^{\alpha_{ic}-1} dp_i \\ &= \sum_{c=1}^C \left[ (y_{ic} - \mathbb{E}[p_{ic}])^2 + \text{Var}(p_{ic}) \right] \\ &= \sum_{c=1}^C \left( y_{ic} - \frac{\alpha_{ic}}{L_i} \right)^2 + \frac{\alpha_{ic}(L_i - \alpha_{ic})}{L_i^2(L_i + 1)},\end{aligned}\quad (4)$$

where  $\mathbb{E}[p_{ic}]$  is estimated by  $\frac{\alpha_{ic}}{L_i}$ , and  $\text{Var}(p_{ic})$  denotes the expected value and variance, respectively. For clarity, we denote the least-squares loss  $\mathcal{L}_{ls}(\alpha_i, y_i^s)$  of the  $i$ -th sample as  $\mathcal{L}_i$ .

The entire source samples are divided into a clean set and a noisy set based on the least-squares loss (Equation (4)). Inspired by the memorization effect of deep neural networks (DNNs), clean (easy) data usually exhibit smaller loss values than noisy (difficult) data in the early stages of training. Therefore, we adopt the entire LS loss  $\{\mathcal{L}_i\}_{i=1}^{N_s}$  to fit the Gaussian mixture model through the Expectation-Maximization algorithm, and compute the posterior probability  $p(z|\mathcal{L}_i) = p(z)p(\mathcal{L}_i|z)/p(\mathcal{L}_i)$ . The posterior probability  $p(z|\mathcal{L}_i)$  means the probability that the  $i$ -th sample is clean or noisy, where  $z \in \{0, 1\}$  denotes whether it belongs to the clean or noisy component.

**Uncertainty-Guided Collaborative Partitioning (UCP).** Extensive experiments reveal that relying solely on posterior probability is vulnerable to estimation bias in the early training stage, which can result in certain noisy samples being mistakenly classified as clean. Therefore, we incorporate both posterior probability  $p(z|\mathcal{L}_i)$  and uncertainty  $u_i$  as dual constraints for clean sample selection (i.e., selecting samples with high posterior probability and low uncertainty), which substantially reduces the risk of mistakenly including noisy samples. More specifically, we set a threshold  $\epsilon$  on  $w_i = p(z=0|\mathcal{L}_i)$  and a threshold  $\delta$  on the uncertainty  $u_i$  to partition the entire source data into clean sample subset  $\mathcal{D}_c$  and noisy sample subset  $\mathcal{D}_n$ , which is formulated as follows:

$$\mathcal{D}_c = \{(x_i^s, y_i^s) | (w_i > \epsilon) \cap (u_i < \delta), \forall (x_i^s, y_i^s) \in \mathcal{D}_s\}, \quad (5)$$

$$\mathcal{D}_n = \{(x_i^s, y_i^s) | (w_i \leq \epsilon) \cup (u_i \geq \delta), \forall (x_i^s, y_i^s) \in \mathcal{D}_s\}. \quad (6)$$

**Noise-Resilient Label Correction.** To prevent the model from being misled by incorrect labels and to expand the pool of reliable samples, we propose Noise-Resilient Label Correction (NRLC) for noisy label samples in  $\mathcal{D}_n$ , as computed in Equation (6). More specifically, for each noise sample  $x_i^s \in \mathcal{D}_n$ , we first identify its  $k$  clean neighbors from the clean subset  $\mathcal{D}_c$  using the  $k$ -nearest neighbor algorithm, denoted as  $\mathcal{N}(x_i^s, k)$ . These neighbors provide a reference distribution for label correction. To assess the consistency between the model's prediction  $\hat{y}_i^s$  and its neighbors' true labels  $y_j^s$ , we compute the Jensen-Shannon divergence (JSD) for each neighbor  $\mathcal{S}(x_i^s) = \sum_{j=1}^k \text{JSD}(\hat{y}_i^s, y_j^s)$ , where smaller  $\mathcal{S}(x_i^s)$  indicates higher consistency. Only samples satisfying  $\mathcal{S}(x_i^s) < \gamma$  are considered for

pseudo-label correction. For these selected samples, we generate corrected labels via a weighted strategy that emphasizes neighbors with distributions more consistent with the model prediction:

$$\tilde{y}_i^s = \arg \max_c \sum_{j=1}^k (1 - \text{JSD}(\hat{y}_i^s, y_j^s)) \cdot y_j^s, \quad c = 1, 2, \dots, C. \quad (7)$$

Thus, each noisy sample is assigned a corrected pseudo-label ( $x_i^s, \tilde{y}_i^s$ ), providing a more reliable supervision signal for subsequent training and enhancing the model's robustness to label noise.

### 3.3 Progressive Cross-Domain Alignment

**Category-Level Contrastive Learning.** Previous studies [4] have demonstrated that constructing contrastive learning between prototypes of the same category from the source and target domains can alleviate the category conflict issue inherent in instance-level methods. Motivated by this, we introduce Category-Level Contrastive Learning (CCL), which explicitly aligns the category-level semantic structures of the source and target domains in the shared feature space. Specifically, we treat prototype pairs belonging to the same category across domains as positive pairs, while those from different categories are regarded as negative pairs, thereby promoting both consistency and discriminability of cross-domain representations. Category-level contrastive learning is determined by the following process:

$$\mathcal{L}_{ccl} = \frac{1}{C} \sum_{c=1}^C -\log \frac{\exp(\frac{(p_c^s)^T p_c^t}{\tau})}{\exp(\frac{(p_c^s)^T p_c^t}{\tau}) + \sum_{i \neq c} \exp(\frac{(p_c^s)^T p_i^t}{\tau})}, \quad (8)$$

where  $p_c^s$  represents the prototype representation of the  $c$ -th class, which is obtained by the mean value of samples with source domain label  $c$ . In each successive epoch,  $p_c^s$  is updated based on the sampled data. For the unlabeled target sample  $x_i^t$ , we first assign the target domain pseudo-label:

$$\hat{y}_i^t = \arg \max_c \cos(f(x_i^t), p_c^s). \quad (9)$$

Subsequently, the target domain prototype  $p_c^t$  is obtained by applying the same process as the source domain prototype. At this point, not surprisingly, we are able to perform class-level contrastive loss, laying the foundation for subsequent hash code alignment learning. Note that this pseudo-label assignment mechanism is sensitive to the quality of the source domain prototype and domain shift, which means that target domain pseudo-labels are prone to assignment bias, i.e., being incorrectly assigned to representations that do not belong to the same source domain prototype. To overcome the aforementioned shortcomings, we replace the target domain pseudo-label generation in Equation (9) with Equation (12) to improve the quality of target pseudo-labels.

**Progressive Partial Optimal Transport.** Assigning target domain pseudo-labels via an optimal transport schedule remains challenging. Conventional Optimal Transport (OT) [36] imposes  $(\|\mu\|_1 = \|v\|_1)$  and requires that all unlabeled probability masses be transported. While this assumption works well for tasks involving the same distribution, it encounters challenges in transferring between  $\mu$  and  $v$  with a large domain shift, where only a portion of samples in  $\mu$  align with  $v$ . However, each unlabeled target domain sample is forcibly transported to a specific source domain prototype,

which disrupts the efficient transport of aligned samples from  $\mu$  to  $v$ . To overcome these drawbacks, we propose a progressive alignment strategy (PPOT) based on Partial Optimal Transport [1, 5].

Formally, the PPOT aims to transport a specified fraction of mass  $0 \leq \tau \leq \min(\|\mu\|_1, \|v\|_1)$  between two distributions at a minimal cost. The set of admissible couplings  $U_\tau(\mu, v)$  is defined as:

$$U_\tau(\mu, v) = \left\{ \pi \in \mathbb{R}_+^{n \times m} \mid \pi \mathbf{1}_m \leq \mu, \pi^T \mathbf{1}_n \leq v, \mathbf{1}_n^T \pi \mathbf{1}_m = \tau \right\}. \quad (10)$$

Therefore, the PPOT problem is defined as:

$$\min_{\pi \in U_\tau(\mu, v)} \langle \pi, Q \rangle_F = \min_{\pi \in U_\tau(\mu, v)} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} Q_{ij}, \quad (11)$$

where the transport cost denotes  $Q = FP$ ,  $P$  represents the prototype of the source domain and  $F$  represents the target domain features. Subsequently, we use PPOT to acquire pseudo-labels  $\hat{y}_i^t \in Y$  for each target domain sample, where  $Y$  is the set of all pseudo-labels, which is formulated as follows:

$$\hat{y}_i^t = \arg \max_{j \in Y} \pi_{ij}^*(\mu, v, \tau). \quad (12)$$

During the PPOT algorithm's target domain pseudo-label assignment process, the cost matrix quantifies the cost of aligning source and target domain samples. This matrix indicates that the more similar a sample  $x_i^t$  is to its prototype  $p_c^s$ , the lower the cost of aligning them. By minimizing the transfer plan  $\pi$ , the PPOT algorithm tends to assign the target sample's mass to the most similar source domain prototype, thereby allocating pseudo-labels in an easy-to-hard order.

**Adversarial Alignment Learning.** Domain-adversarial alignment introduces adversarial training between a domain discriminator and a feature extractor, encouraging the source and target features to become indistinguishable in a shared representation space. This process facilitates the learning of domain-invariant features and effectively reduces the distributional discrepancy between the two domains. Specifically, the adversarial constraint is as follows:

$$\mathcal{L}_{aal}(\mathcal{D}_s, \mathcal{D}_t) = -\frac{1}{N_s} \sum_{i=1}^{N_s} \log(d(f(x_i^s))) - \frac{1}{N_t} \sum_{i=1}^{N_t} \log(1 - d(f(x_i^t))), \quad (13)$$

where  $d$  denotes the domain discriminator and  $f$  denotes the feature extractor. By leveraging a gradient reversal layer, the feature extractor seeks to confuse the domain discriminator by producing domain-invariant features, whereas the discriminator strives to distinguish features from different domains, thus globally aligning the source data with the target data in the feature space.

**Optimization.** To learn domain-invariant discriminative hashing representations across various domains, the commonly utilized relation preserving loss  $\mathcal{L}_{rel}$  [4, 20] and quantization loss  $\mathcal{L}_q$  [26] are also adopted along with the noise separation loss  $\mathcal{L}_{ls}$  and cross-domain alignment loss  $\mathcal{L}_{ccl} + \mathcal{L}_{aal}$  to optimize. Thus, the overall training objective is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{ls} + \lambda_1 \mathcal{L}_{ccl} + \lambda_2 \mathcal{L}_{aal} + \lambda_3 \mathcal{L}_q + \lambda_4 \mathcal{L}_{rel}, \quad (14)$$

where  $\lambda_1, \lambda_2, \lambda_3$  and  $\lambda_4$  are trade-off hyper-parameters during the training phase. The quantization loss  $\mathcal{L}_q$  ensures that the relaxed hash code  $H$  approximates the binary hash code  $B$ , where  $B = \text{sign}(H)$ , and  $H$  is the final hashing network output. The relation preserving loss  $\mathcal{L}_{rel}$  maintains the similarity between samples in this feature space into the hash codes.

## 4 Experiments

To comprehensively evaluate the performance of our proposed ELPA, we conduct extensive experiments on three widely used benchmark datasets, briefly introduced below.

**Datasets: Office-31 [29]:** comprises three distinct domains: Amazon (A), Webcam (W), and DSLR (D). It contains 4,110 images across 31 different categories. Following prior studies, we evaluate six transfer tasks:  $A \rightarrow W$ ,  $W \rightarrow A$ ,  $A \rightarrow D$ ,  $D \rightarrow A$ ,  $W \rightarrow D$ , and  $D \rightarrow W$ .

**Office-Home [35]:** covers four different domains: Product images (Pr), Real-World images (Rw), Artistic images (Ar), and Clip Art (Cl), totaling 15,500 images from 65 categories. Consistent with previous works, we evaluate six cross-domain retrieval tasks:  $Pr \rightarrow Rw$ ,  $Rw \rightarrow Pr$ ,  $Rw \rightarrow Ar$ ,  $Ar \rightarrow Rw$ ,  $Cl \rightarrow Ar$ , and  $Ar \rightarrow Cl$ .

**Digits:** This benchmark includes two digit datasets, MNIST [19] and USPS [14]. Each can serve as either the source or target domain, leading to two transfer tasks: MNIST (M)  $\rightarrow$  USPS (U) and USPS (U)  $\rightarrow$  MNIST (M).

**Evaluation Metrics:** We conduct a comprehensive evaluation using three widely adopted metrics: mean Average Precision (mAP), Precision-Recall (P-R) curve, and Top-N Precision curve.

**Baselines:** In terms of domain-adaptive hash baselines, we select a wide range of state-of-the-art baseline methods for comparison: ITQ [8], GTH-g [22], GTH-h [22], DAPH [12], PWCF [13], PEACE [37], ROSE [42], CPH [4], and COUPLE [25].

**Implementation Details:** All experiments are conducted in PyTorch using a single RTX 3090 GPU. We optimize the proposed ELPA using mini-batch stochastic gradient descent (SGD) with momentum. The batch size is 256, the learning rate is set to 0.0001, and the total number of training epochs is 500. Furthermore, we use VGG-16 as the backbone network for feature extraction, and construct a two-layer MLP hash encoder on its output to map visual features into Hamming space. For the hyper-parameters of ELPA, according to sensitivity analysis, the hyper-parameters  $\epsilon$  and  $\delta$  are set to 0.1 and 0.8, respectively. For the trade-off hyper-parameters  $\lambda_1, \lambda_2, \lambda_3$ , and  $\lambda_4$  in the overall optimization objective, we set (1)  $\lambda_1 = 0.1, \lambda_2 = 0.01, \lambda_3 = 0.1, \lambda_4 = 0.1$ ; (2)  $\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = 100, \lambda_4 = 10$ ; (3)  $\lambda_1 = 0.01, \lambda_2 = 0.01, \lambda_3 = 0.01, \lambda_4 = 0.01$  on the Office-31, Office-Home, and Digits datasets. The remaining parameters are fixed at  $\tau = 0.8, \tau' = 0.5$ , and the number of warm-up epochs is 5, 5, and 15 for the three datasets, respectively. Each dataset contains 10% of the target samples for testing, and the remaining samples are used for training according to [4, 13]. In the cross-domain retrieval task, the query comes from the target domain and the gallery comes from the source domain. In the single-domain retrieval task, the query and gallery come from the target domain.

### 4.1 Experiment Results and Analysis

**Results on Cross-Domain Retrieval Tasks:** Table 1 reports the mAP metric of all methods on the cross-domain retrieval task. Our

**Table 1: Comparison of cross-domain retrieval performance against baseline methods on the Office-31 and Office-Home benchmarks under varying noise ratios. The best results are shown in bold, and the second-best are underlined.**

| Noise | Methods | Office-31    |              |              |              |              |              | Office-Home  |              |              |              |              |              | Avg.         |
|-------|---------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|       |         | A → WW       | → AA         | → DD         | → AW         | → DD         | → W          | Pr → Rw      | Rw → Pr      | Rw → Ar      | Ar → Rw      | Cl → Ar      | Ar → Cl      |              |
| 40%   | ITQ     | 22.03        | 22.47        | 25.20        | 27.12        | 54.48        | 45.42        | 35.97        | 35.91        | 28.09        | 27.96        | 13.96        | 12.85        | 29.29        |
|       | GTH-g   | 9.71         | 9.37         | 10.70        | 9.90         | 20.49        | 22.21        | 12.64        | 13.59        | 11.04        | 11.70        | 4.92         | 5.40         | 11.81        |
|       | GTH-h   | 11.74        | 9.64         | 13.91        | 9.53         | 18.47        | 18.45        | 13.91        | 12.75        | 10.75        | 11.37        | 5.05         | 5.78         | 11.78        |
|       | PWCF    | 10.80        | 11.81        | 15.39        | 10.13        | 18.89        | 20.22        | 16.00        | 18.77        | 16.98        | 13.97        | 7.75         | 3.93         | 13.72        |
|       | DAPH    | 12.99        | 11.51        | 13.13        | 10.81        | 20.56        | 20.28        | 15.36        | 16.13        | 14.30        | 12.29        | 6.44         | 6.15         | 13.33        |
|       | PEACE   | 27.14        | 33.67        | 41.47        | 38.37        | 82.65        | 64.55        | 50.01        | <u>64.98</u> | 36.85        | <u>56.01</u> | 8.95         | 15.26        | 43.33        |
|       | ROSE    | 43.74        | 29.35        | 47.38        | 28.67        | 45.15        | 41.90        | 20.52        | <u>16.64</u> | 20.66        | 20.58        | 17.56        | 11.58        | 28.64        |
|       | CPH     | <u>50.08</u> | <u>50.97</u> | <u>57.51</u> | <u>49.33</u> | <u>88.98</u> | <u>77.17</u> | <u>55.04</u> | 53.28        | <u>38.06</u> | 43.02        | <u>29.76</u> | <u>19.07</u> | <u>51.02</u> |
|       | COUPLE  | 36.24        | 34.45        | 39.88        | 37.46        | 78.96        | 65.79        | 43.99        | 46.51        | 32.81        | 36.74        | 22.18        | 14.08        | 40.76        |
|       | Ours    | <b>60.05</b> | <b>59.20</b> | <b>68.14</b> | <b>58.05</b> | <b>92.84</b> | <b>82.43</b> | <b>72.55</b> | <b>69.66</b> | <b>57.87</b> | <b>58.65</b> | <b>39.45</b> | <b>23.50</b> | <b>61.87</b> |
| 60%   | ITQ     | 23.06        | 21.59        | 24.87        | 24.08        | 58.53        | 48.07        | 34.53        | 36.19        | 28.10        | 28.33        | 13.87        | 11.79        | 29.42        |
|       | GTH-g   | 6.29         | 6.85         | 6.26         | 6.79         | 10.68        | 10.47        | 6.87         | 7.50         | 6.29         | 5.95         | 3.07         | 3.05         | 6.67         |
|       | GTH-h   | 7.10         | 6.25         | 8.03         | 6.23         | 10.57        | 10.41        | 6.99         | 7.76         | 7.68         | 6.29         | 3.30         | 3.49         | 7.01         |
|       | PWCF    | 9.10         | 6.65         | 9.34         | 6.32         | 12.20        | 8.53         | 9.14         | 11.26        | 11.34        | 7.04         | 4.79         | 2.35         | 8.17         |
|       | DAPH    | 7.54         | 7.43         | 7.68         | 8.55         | 11.01        | 12.61        | 8.09         | 8.76         | 8.49         | 8.53         | 3.57         | 3.25         | 7.96         |
|       | PEACE   | 11.62        | 19.23        | 10.63        | 20.49        | 70.10        | 45.07        | 35.20        | <u>57.10</u> | <u>28.57</u> | <u>45.49</u> | 5.89         | 10.75        | 30.01        |
|       | ROSE    | <u>38.29</u> | 24.03        | 38.17        | 20.14        | 36.83        | 34.20        | 18.92        | 16.44        | 18.26        | 20.93        | 16.21        | 10.42        | 24.40        |
|       | CPH     | <u>37.72</u> | <u>40.59</u> | <u>43.08</u> | <u>37.44</u> | <u>75.90</u> | <u>61.01</u> | <u>39.59</u> | 38.37        | 28.18        | 26.55        | <u>18.93</u> | <u>12.47</u> | <u>38.32</u> |
|       | COUPLE  | 28.90        | 26.53        | 31.87        | 30.65        | 69.47        | 58.16        | 30.30        | 32.20        | 25.85        | 23.23        | 13.55        | 9.96         | 31.72        |
|       | Ours    | <b>55.06</b> | <b>56.02</b> | <b>62.73</b> | <b>49.62</b> | <b>87.76</b> | <b>72.61</b> | <b>67.41</b> | <b>66.54</b> | <b>55.54</b> | <b>46.98</b> | <b>32.61</b> | <b>20.80</b> | <b>56.14</b> |
| 80%   | ITQ     | 21.95        | 21.24        | 24.58        | 26.71        | 58.89        | 47.25        | <u>34.50</u> | <u>36.62</u> | <u>28.55</u> | <u>26.82</u> | <u>13.61</u> | <u>12.42</u> | 29.43        |
|       | GTH-g   | 4.60         | 4.40         | 4.68         | 4.36         | 5.68         | 6.42         | 3.13         | 3.24         | 2.97         | 2.69         | 2.17         | 2.16         | 3.88         |
|       | GTH-h   | 4.70         | 4.57         | 4.85         | 4.90         | 6.29         | 6.38         | 3.02         | 3.29         | 2.95         | 2.88         | 2.19         | 2.26         | 4.02         |
|       | PWCF    | 4.72         | 4.95         | 4.20         | 4.81         | 5.96         | 5.27         | 3.29         | 3.42         | 3.28         | 3.18         | 2.45         | 2.33         | 3.99         |
|       | DAPH    | 4.56         | 4.29         | 5.01         | 4.64         | 5.96         | 6.59         | 3.29         | 3.42         | 3.28         | 3.18         | 2.44         | 2.33         | 4.08         |
|       | PEACE   | 7.97         | 11.73        | 5.54         | 9.03         | 31.43        | 18.44        | 10.94        | 31.19        | 6.95         | 23.77        | 4.03         | 3.68         | 13.73        |
|       | ROSE    | 23.47        | 18.64        | 23.81        | 17.33        | 27.94        | 30.72        | 11.86        | 13.25        | 12.96        | 11.71        | 11.75        | 7.46         | 17.58        |
|       | CPH     | <u>28.39</u> | <u>29.54</u> | <u>32.35</u> | <u>31.77</u> | <u>62.73</u> | <u>51.64</u> | 28.94        | 32.14        | 23.40        | 21.48        | 12.64        | 9.22         | <u>30.35</u> |
|       | COUPLE  | 26.17        | 24.10        | 29.47        | 29.47        | 60.63        | 50.85        | 27.15        | 29.55        | 23.59        | 19.32        | 11.76        | 9.28         | 28.45        |
|       | Ours    | <b>47.36</b> | <b>39.88</b> | <b>52.75</b> | <b>37.67</b> | <b>72.93</b> | <b>60.51</b> | <b>56.85</b> | <b>50.15</b> | <b>43.35</b> | <b>32.69</b> | <b>22.51</b> | <b>18.08</b> | <b>44.56</b> |

**Table 2: Comparison of single-domain retrieval mAP results on MNIST and USPS datasets under 40% label noise.**

| Methods | MNIST → USPS |              |              |              |              |
|---------|--------------|--------------|--------------|--------------|--------------|
|         | 16           | 32           | 64           | 128          | Avg.         |
| CPH     | 58.08        | 59.74        | 62.20        | 63.66        | 60.92        |
| COUPLE  | 27.92        | 29.57        | 27.56        | 30.13        | 28.80        |
| Ours    | <b>64.70</b> | <b>66.74</b> | <b>68.76</b> | <b>69.01</b> | <b>67.30</b> |
| Methods | USPS → MNIST |              |              |              |              |
|         | 16           | 32           | 64           | 128          | Avg.         |
| CPH     | 35.62        | 39.86        | 44.75        | 49.15        | 42.35        |
| COUPLE  | 27.39        | 25.79        | 26.30        | 30.22        | 27.43        |
| Ours    | <b>45.00</b> | <b>51.90</b> | <b>52.05</b> | <b>53.33</b> | <b>50.57</b> |

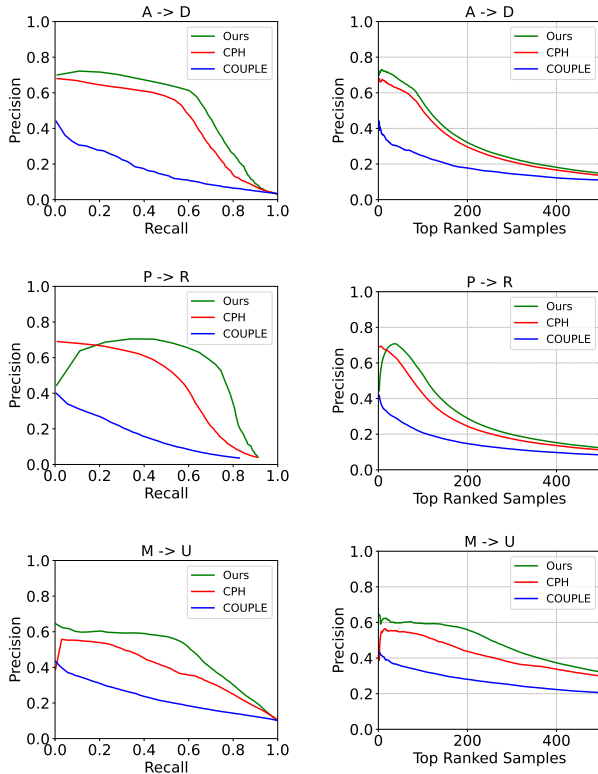
method ELPA significantly outperforms other methods across all noise rates on almost all datasets. Furthermore, when the noise ratio is extremely high, our method is more effective than baseline methods in mitigating the noise-induced degradation of hash retrieval performance. For instance, on the Office-31 and Office-Home datasets with 60% noisy labels, ELPA surpasses the state-of-the-art

method CPH by an average of 17.82%. Even under 40% noise, our method still achieves improvements of 10.85%. This shows that CPH and other comparison methods cannot effectively learn the shared discriminative hash representation when encountering noisy label pollution. Our method effectively learns compact and discriminative hash codes while mitigating noise, thanks to ELPA’s powerful noise modeling and robust cross-domain alignment.

**Results on Single-Domain Retrieval Tasks:** We selected the Digits dataset to verify the effectiveness and generalization ability of our proposed ELPA method in single-domain retrieval. Table 2 shows the single-domain mAP results on the Digits dataset with hash code lengths ranging from 16 to 128 bits. The quantitative results in Table 2 show that ELPA achieves competitive results in both M→U and U→M, achieving the highest mAP scores in retrieval with different hash code lengths, reaching 64.70%, 66.74%, 68.76%, and 69.01%, and 45.00%, 51.90%, 52.05%, and 53.33%, respectively. Moreover, compared with the second-best result obtained by CPH, our method achieves average improvements of more than 6.38% and 8.22%, respectively. These mAP improvements strongly demonstrate that our ELPA can effectively mitigate the impact of

**Table 3: Ablation results on the Office-Home dataset with 80% label noise. The best performance is highlighted in bold.**

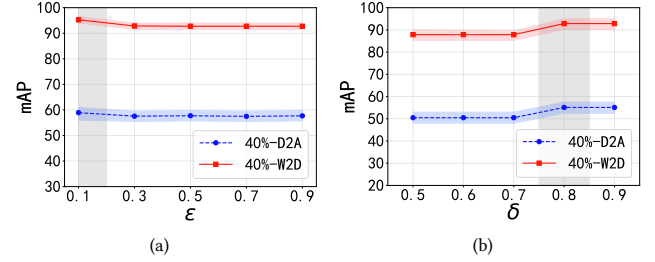
| No. | UNS | ENS | PPOT | $\mathcal{L}_{ccl}$ | $\mathcal{L}_{aal}$ | Pr $\rightarrow$ Rw | Rw $\rightarrow$ Pr | Rw $\rightarrow$ Ar | Ar $\rightarrow$ Rw | Cl $\rightarrow$ Ar | Ar $\rightarrow$ Cl | Avg.         |
|-----|-----|-----|------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------|
| 1   | ✓   |     | ✓    | ✓                   | ✓                   | <b>56.85</b>        | <b>50.15</b>        | <b>43.35</b>        | <b>32.69</b>        | <b>22.51</b>        | <b>18.08</b>        | <b>37.27</b> |
| 2   |     | ✓   | ✓    | ✓                   | ✓                   | 48.45               | 48.85               | 42.36               | 30.24               | 20.97               | 16.14               | 34.50        |
| 3   | ✓   |     |      | ✓                   | ✓                   | 49.74               | 48.93               | 42.19               | 30.86               | 21.90               | 16.01               | 34.94        |
| 4   |     |     | ✓    | ✓                   | ✓                   | 33.13               | 30.30               | 25.50               | 21.57               | 14.56               | 11.92               | 22.83        |
| 5   | ✓   |     | ✓    |                     | ✓                   | 46.56               | 48.02               | 41.89               | 27.22               | 15.61               | 13.35               | 32.11        |
| 6   | ✓   |     | ✓    | ✓                   |                     | 50.11               | 48.80               | 42.16               | 28.40               | 21.12               | 14.01               | 34.10        |

**Figure 2: Comparison of 64-bit Precision-Recall and Top-N precision curves of different methods on Office-31, Office-Home, and Digits with 40% noise.**

noise on single-domain retrieval performance. Figure 2 evaluates single-domain and cross-domain retrieval performance using both precision-recall and Top-K precision curves. The results demonstrate ELPA’s dual advantages: on the precision-recall curve, ELPA’s higher coverage demonstrates its robustness against noise; on the Top-K precision curve, its performance continues to lead as the value of K increases, demonstrating the model’s powerful ability to accurately rank and prioritize relevant examples. These results consistently demonstrate the effectiveness and superiority of ELPA.

## 4.2 Ablation Study

As shown in Table 3, we conduct a fair ablation study to comprehensively evaluate the contributions of the proposed components.

**Figure 3: Parameter sensitivity analysis of  $\epsilon$  and  $\delta$  on the Office-31 dataset with 40% label corruption.**

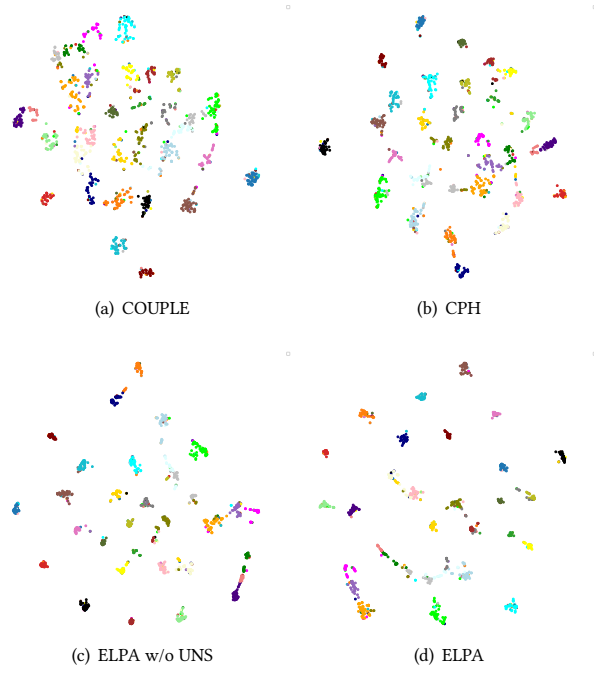
These components were extensively evaluated on Office-Home with 80% noise, while removing one or more components.

To demonstrate the effectiveness of our proposed Uncertainty-guided Collaborative Partitioning (UCP) for noise detection, we compared the uncertainty-aware noise separation module (UNS) with the noise partitioning obtained using only the Evidential Learning for Noise Separation (ENS) (No. 1 vs. No. 2). The UCP improves the average mAP of the ENS module by 2.77%. This is attributed to the fact that uncertainty provides intrinsic evidence mass for noise separation, thereby effectively identifying potentially high-loss and low-uncertainty noisy samples. Furthermore, the proposed UNS module (No. 1 vs. No. 4) contributes most significantly to ELPA, achieving an average performance gain of 14.44%. This indicates that accurately and reliably dividing the corrupted dataset into different subsets can effectively mitigate the impact of the noisy label on domain-adaptive hash. In addition, the original target domain pseudo-label generation is replaced by the Progressive POT component (PPOT) for comparison (No. 1 vs. No. 3). In comparison, the Progressive POT module improves the average mAP performance of the Office-Home dataset by 2.33%. These results show that the Progressive POT can effectively alleviate the pseudo-label misassignment problem under domain shift and noisy labels.

To evaluate  $\mathcal{L}_{ccl}$  and  $\mathcal{L}_{aal}$  loss, we remove  $\mathcal{L}_{ccl}$  and  $\mathcal{L}_{aal}$  from our model respectively. Table 3 shows that ELPA outperforms ELPA (without  $\mathcal{L}_{ccl}$ ) by an average of 5.16% and ELPA (without  $\mathcal{L}_{aal}$ ) by an average of 3.17% on the Office-Home dataset. This indicates that class-level contrastive learning and adversarial alignment learning are effective for learning cross-domain hash codes.

## 4.3 Parameter Sensitivity

To explore the variations of mAP with changing hyper-parameter values, we conduct parameter sensitivity analysis experiments about hyper-parameters  $\epsilon$  and  $\delta$  in Equation (5) on the Office-31

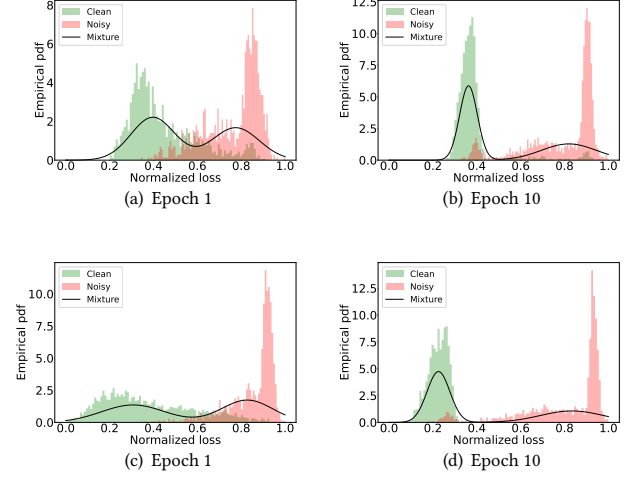


**Figure 4: The t-SNE visualization comparison of different methods on the Office-31 dataset with 40% label noise.**

dataset under the 40% noise ratio, where  $\epsilon$  and  $\delta$  are the GMM discrimination threshold and uncertainty threshold for noise sample selection, respectively. Specifically, to accurately reflect the performance variations affected by hyper-parameters, we evaluate one of the hyper-parameters, the other hyper-parameters are set to be unchanged, which follows the principle of controlling variables. Figures 3 (a) and (b) show the impact of  $\epsilon$  and  $\delta$  on Office-31 dataset, respectively. It can be seen that mAP decreases with increasing  $\epsilon$  and begins to stabilize after exceeding 0.1. Regarding the hyper-parameter  $\delta$ , we can see that our performance is insensitive to  $\delta$  within the range [0.8, 0.9], but performance degrades beyond this range. Thus, we can conclude that our ELPA achieves better performance by setting  $\epsilon = 0.1$  and  $\delta = 0.8$  with the optimal combination of these hyper-parameters.

#### 4.4 Visualization

To further validate the effectiveness of the proposed method, we conduct a t-SNE visualization of the hash codes generated by different approaches. As shown in Figure 4, the results show that compared to competitive baselines, our method exhibits clearer and more compact clustering boundaries with more pronounced inter-class separation. This observation provides direct evidence that our approach is capable of learning more discriminative and robust hashing representations, thereby maintaining high semantic consistency even under noisy and cross-domain conditions. Moreover, by comparing ELPA and ELPA without UNS (denotes ELPA w/o UNS), we observe that incorporating uncertainty-guided noise separation enables ELPA to further enhance inter-class separability and preserve intra-class compactness. The resulting clear clustering structure and stable class boundaries offer intuitive evidence of our model’s superior performance in noisy cross-domain scenarios.



**Figure 5: We visualize the loss distributions of clean and noisy images at different training stages on Office-31 and Office-Home with 40% noise. Thanks to our uncertainty-aware noise separation module, the loss distribution of clean and noisy samples exhibits distinct polarizations.**

To further analyze and understand the proposed ELPA, we show several qualitative examples on the Office-31 and Office-Home datasets with 40% noise. Figure 5 visualizes the loss distribution of the per-sample during training. It can be observed that after training, the loss distribution of the clean sample becomes distinctly separated from that of noisy samples, manifesting a clear polarization trend. These observations validate the effectiveness of our uncertainty-aware noise separation method. Therefore, the overall evidence learning method can be naturally used to distinguish between noisy and clean samples, significantly enhancing the robustness of cross-domain hashing.

## 5 Conclusion

This paper reveals a common problem in cross-domain hash retrieval that noisy labels and domain shifts complicate effective cross-domain alignment. To tackle these issues, we propose a novel robust cross-domain hashing framework named ELPA. In particular, we integrate the evidential learning for the noise separation module with noise-resilient label correction to effectively mitigate overfitting arising from noisy labels. Additionally, we leverage category-level contrastive learning and progressive partial optimal transport to strengthen cross-domain alignment, thereby alleviating performance degradation induced by domain shifts. Extensive experiments on three benchmark datasets demonstrate the superior performance of the proposed method.

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